

Airlines Flight Data Analysis

Project Report



Project Overview

This project analyzes over 300,000 airline flight records to understand pricing behavior, flight duration patterns, and the impact of class, stops, and routes on ticket prices.



Pricing Behavior

Uncover trends in ticket costs.



Duration Patterns

Analyze flight time influences.



Route Impact

Assess how routes affect prices.

Tools & Technologies

Our analysis leveraged a suite of powerful tools to process and visualize flight data.



Python

For data manipulation (Pandas, NumPy).



Matplotlib

For data visualization.



Jupyter Notebook

For interactive analysis.



CSV Dataset

Raw flight records.

Dataset Description

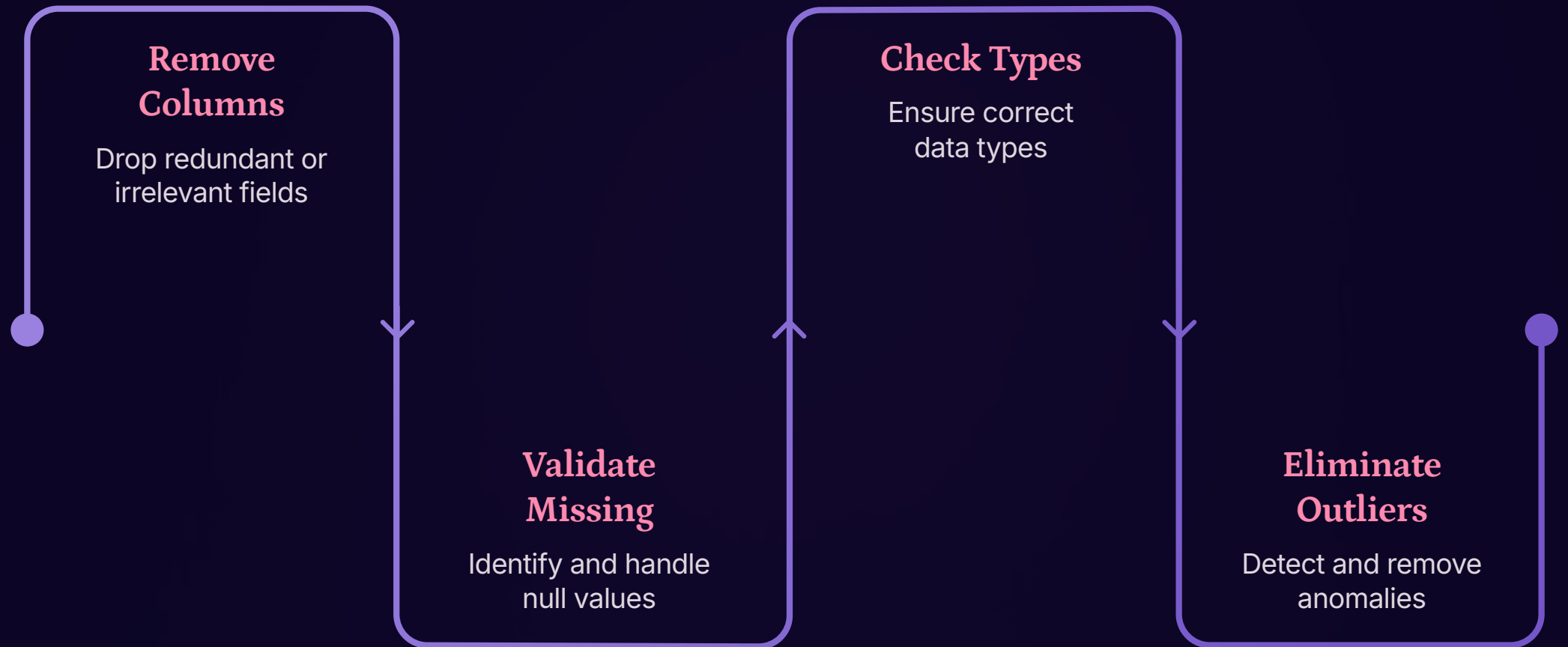


The dataset contains airline name, source and destination cities, class type, flight duration, number of stops, ticket price, and departure/arrival details.

- Over 300,000 records
- Real-world pricing behavior
- Suitable for trend analysis

Data Cleaning Process

Ensuring accurate and reliable analysis through meticulous data preparation.



This critical step guarantees the integrity of our analytical results.

Exploratory Data Analysis: Key Insights

Premium vs. Budget

Airline price comparison revealed distinct pricing structures.

Class Impact

Business class tickets are significantly more expensive than economy.

Non-stop Premium

Non-stop flights generally cost more than multi-stop options.

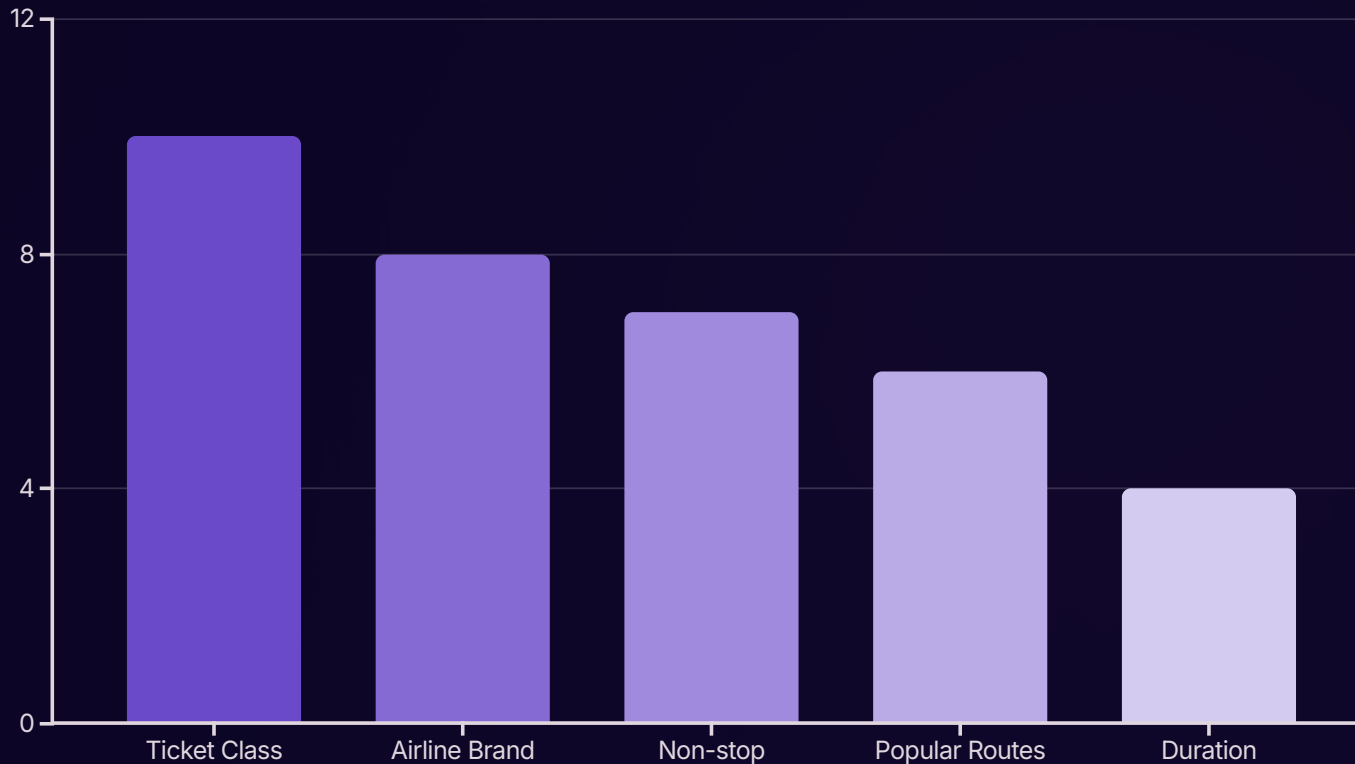
Duration & Brand

Longer flights cost more, but brand and class heavily influence pricing.

Competitive Routes

Popular routes show competitive pricing due to high demand.

Pricing Factors: A Closer Look



Our analysis identified the strongest factors influencing flight ticket prices:

- **Ticket class** is the strongest pricing factor.
- **Airline brand** significantly affects fare structure.
- **Non-stop flights** command premium pricing.
- **Popular routes** increase competition and pricing.
- **Duration** impacts cost but is secondary to other factors.

Key Findings

1 Ticket Class Dominates

The class of travel (economy, business, first) is the primary determinant of ticket price.

2 Brand Value

Established airline brands often charge higher fares due to perceived quality and service.

3 Non-stop Premium

Travelers pay more for the convenience of direct flights.

4 Route Competition

High-demand routes experience competitive pricing dynamics.

Actionable Insights

Data-driven insights can support airline pricing strategies and customer decision-making.



Optimize Pricing

Airlines can adjust fares based on class, stops, and route popularity.



Empower Customers

Travelers can make informed choices by understanding pricing factors.



Predict Trends

Anticipate market shifts and adapt strategies proactively.

Conclusion

This project demonstrates how large-scale flight data can be transformed into actionable insights.

Converting raw data into meaningful decisions for both airlines and travelers.

